Docker

Information security club course II

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v0.1

Outline

1. Container
   - Basic
   - Comparing Containers and Virtual Machines
2. Dockerfile
   - Introduction
   - Write a Dockerfile
   - Docker-compose
3. Docker commands
   - Basic commands
   - Exercises
4. Tiny project
   - Crawler
   - Database
   - Communication
5. Security
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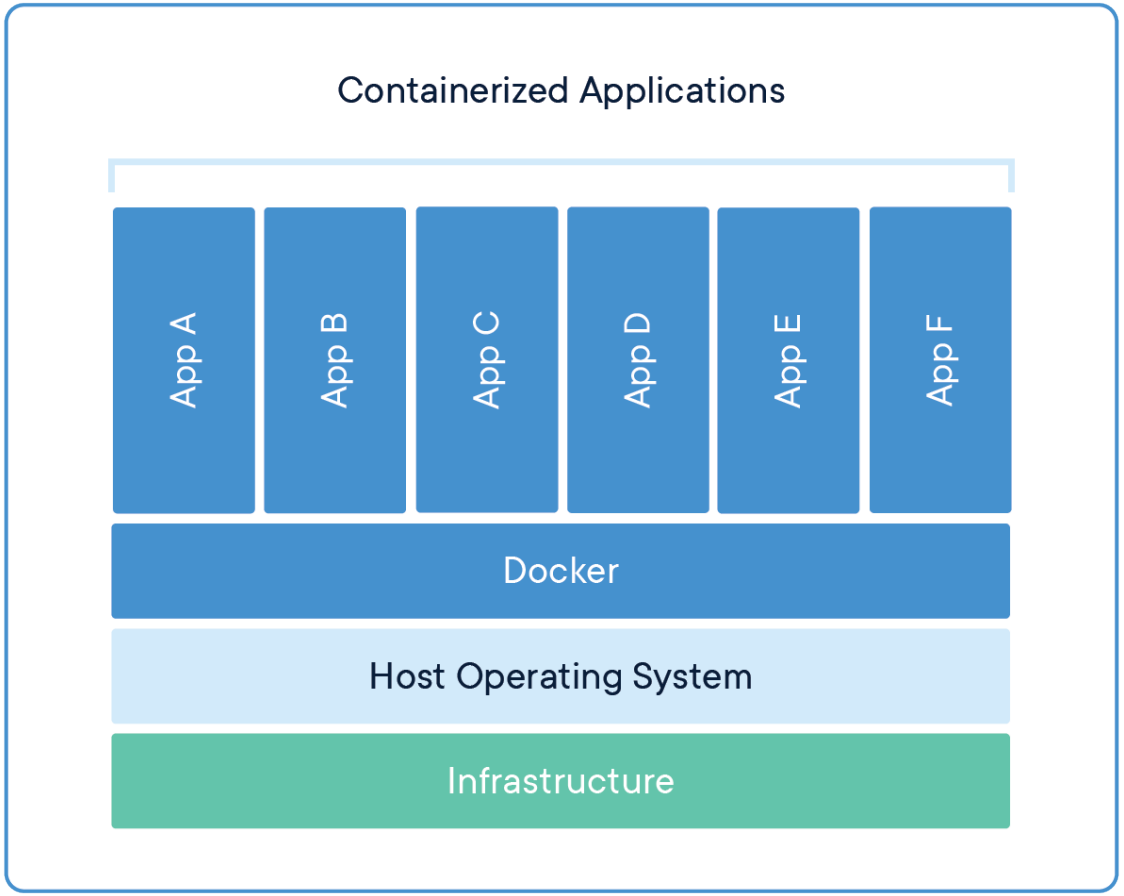
Before the speech

- The 'Learning Corner TA' of operating system course
  - 18:00 → 21:00 (Thur.)
  - EC1013
  - From 22nd March 2021
- Other office hour:
  - 18:00 → 21:00 (Thur.)
  - EC3034
  - Send mail before you come.
- zxc25077667@protonmail.com

Container

Big idea

Containerize

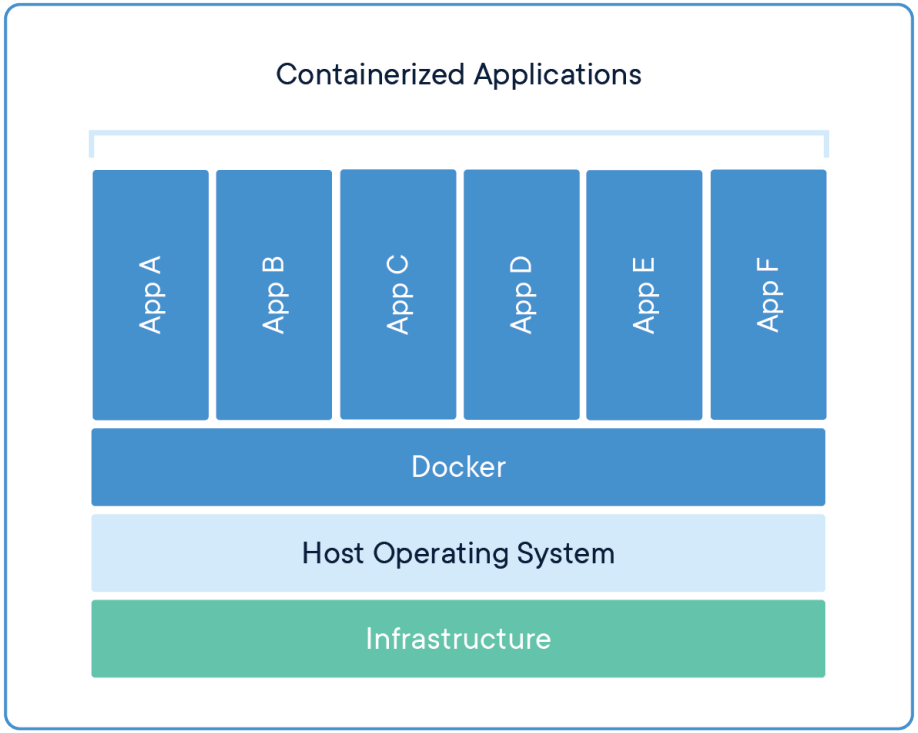


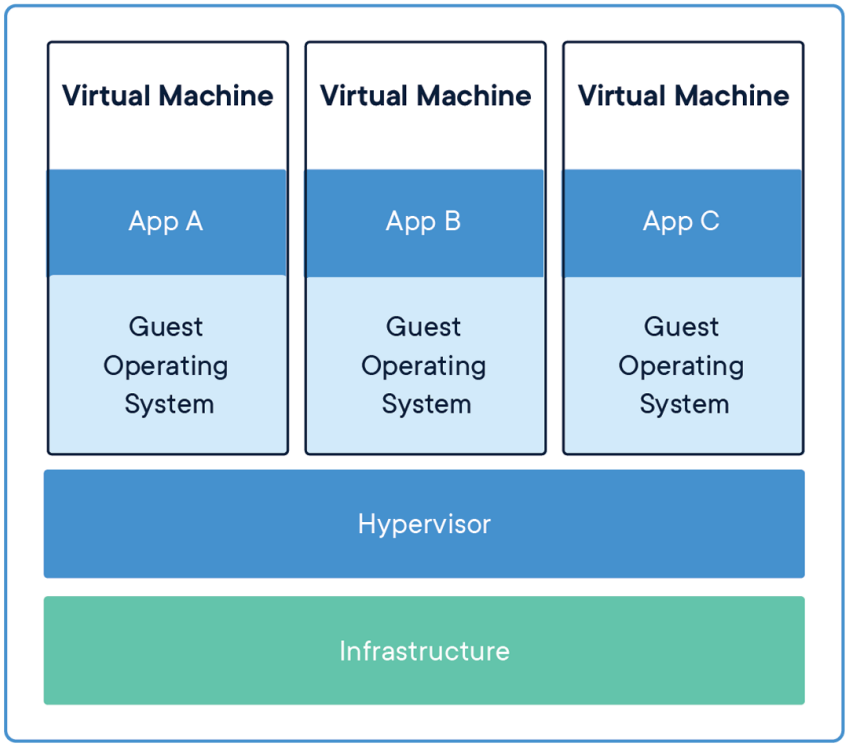
[1]

Big idea

[2]

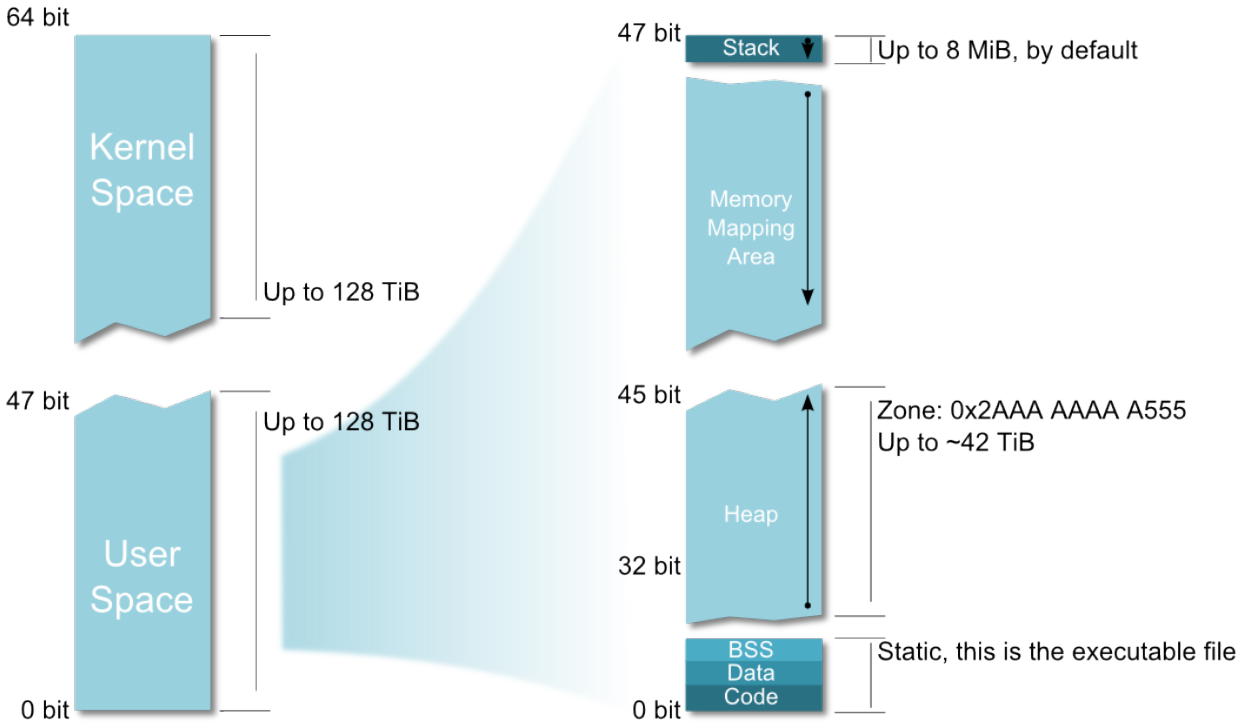
Compare with virtule machines





So, what is share kernel?

Let's recall the OS 101 course

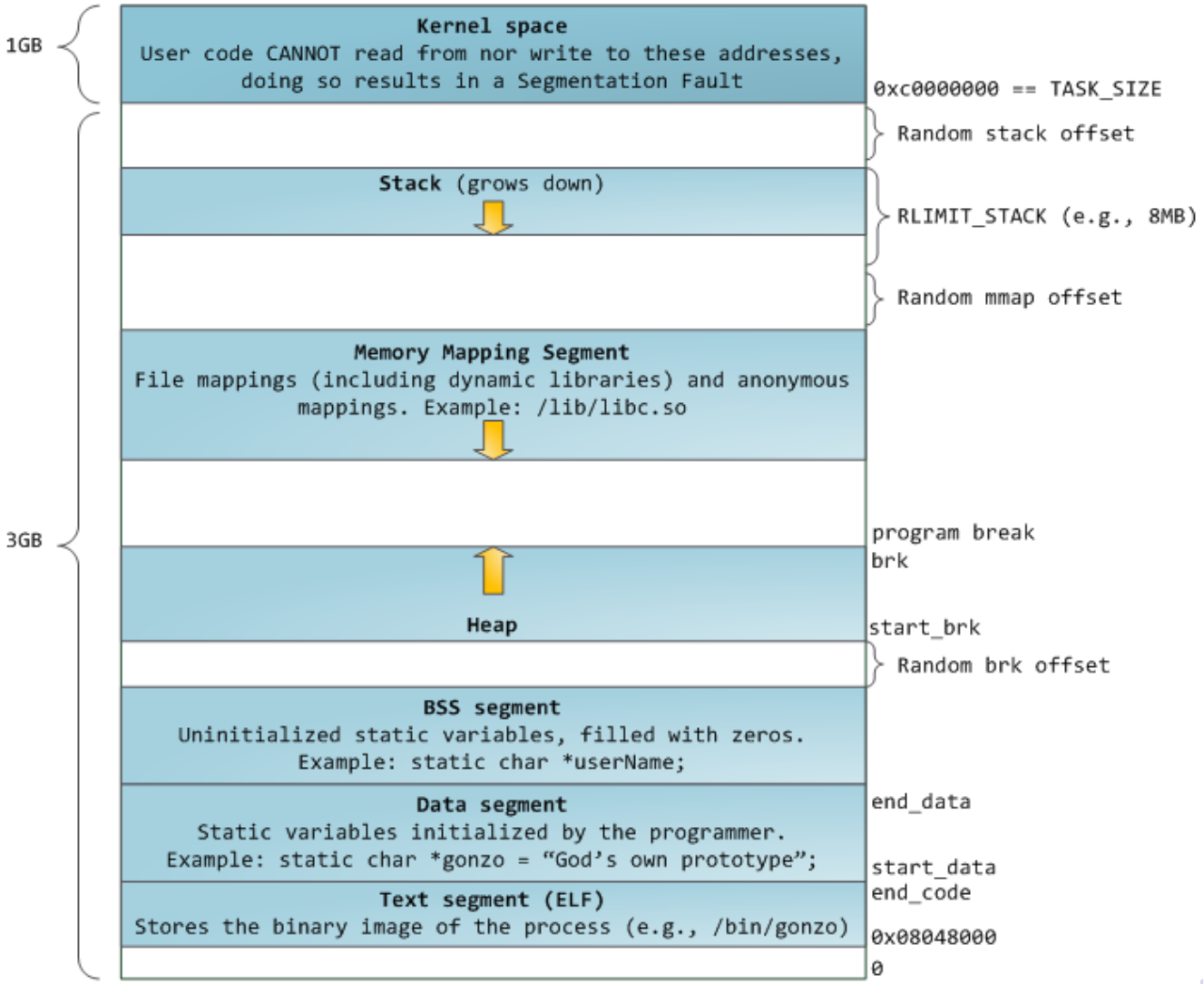


So Kernel + User Spaces add for 256 TiB which is a tiny part of the 16 777 216 TiB addressable over 64 bit!

[3]

So, what is share kernel?

The 32-bits memory layout



What is HYPERVISOR?

Virtual machine monitor

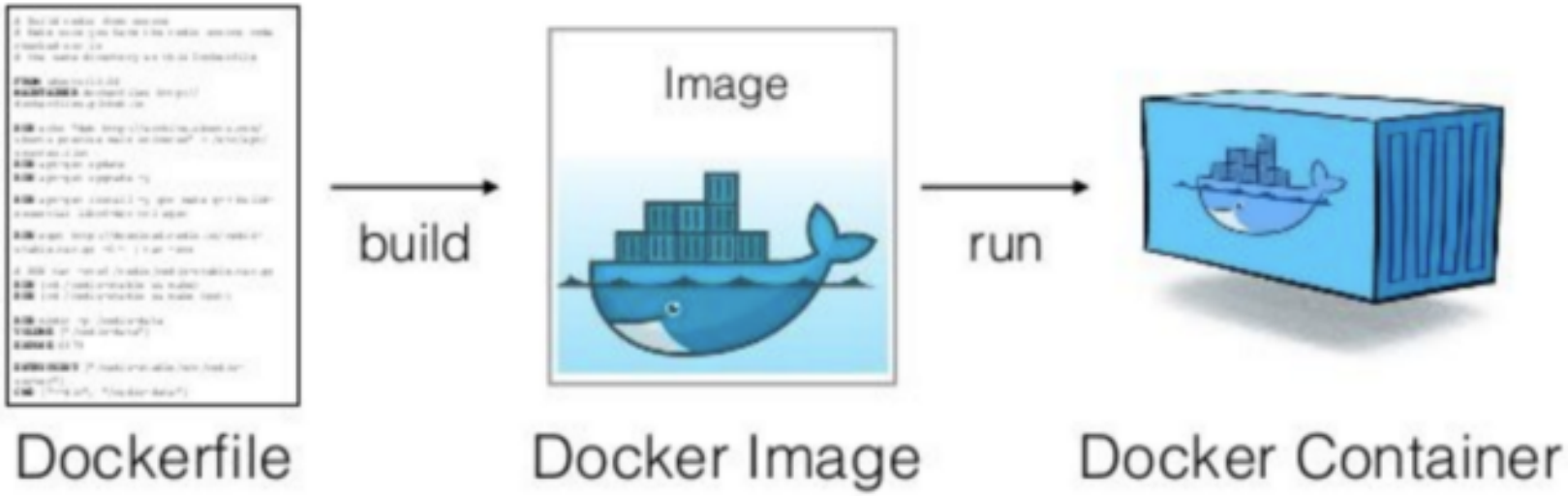
Dockerfile

What is Dockerfile?

Definition:

A text document that contains all the commands a user could call on the command line to assemble an image.

Image and Container



[4]

Like a program in execution is called a process.
An image in execution is called a container.

DockerHub

A public **images** hub. We push/pull **images** from it by default.
We push an image and pull an image rather than a container.

Make sense, right?

Dockerfile 101

Our Dockerfile

Docker-compose 101

Our Docker-compose.yml

Docker commands

Basic

Commands Cheat Sheet

Container Lifecycle

docker create [IMAGE]	create a container without starting it
docker rename [CONTAINER_NAME] [NEW_CONTAINER_NAME]	rename a container
docker run [IMAGE]	create and start a container
docker run --rm [IMAGE]	remove a container after it stops
docker run -td [IMAGE]	start a container and keep it running
docker run -it [IMAGE]	create, start the container, and run a command in it
docker run -it-rm [IMAGE]	create, start the container, and run a command in it; after executing, the container is removed
docker rm [CONTAINER]	delete a container if it isn't running
docker update [CONTAINER]	update the configuration of a container

Image Lifecycle

docker build [URL]	create an image from a Dockerfile
docker build -t [URL]	build an image from a Dockerfile and tags it
docker pull [IMAGE]	pull an image from a registry
docker push [IMAGE]	push an image to a registry
docker import [URL/FILE]	create an image from a tarball
docker commit [CONTAINER] [NEW_IMAGE_NAME]	create an image from a container
docker rmi [IMAGE]	remove an image
docker load [TAR_FILE/STDIN_FILE]	load an image from a tar archieve as stdin
docker save [IMAGE] > [TAR_FILE]	save an image to a tar archive stream to stdout with all parent layers, tags, and versions

Networking

docker network ls	list networks
docker network rm [NETWORK]	remove one or more networks
docker network inspect [NETWORK]	show information on one or more networks
docker network connect [NETWORK] [CONTAINER]	connect a container to a network
docker network disconnect [NETWORK] [CONTINAER]	disconnect a container from a network

Start & Stop

docker start [CONTAINER]	start a container
docker stop [CONTAINER]	stop a running container
docker restart [CONTAINER]	stop a running container and start it up again
docker pause [CONTAINER]	pause processes in a running container
docker unpause [CONTAINER]	unpause processes in a container
docker wait [CONTAINER]	block a container until other containers stop
docker kill [CONTAINER]	kill a container by sending SIGKILL to a running container
docker attach [CONTAINER]	attach local standard input, output, and error streams to a running container

Information

docker ps	list running containers
docker ps -a	list running and stopped containers
docker logs [CONTAINER]	list the logs from a running container
docker inspect [OBJECT_NAME/ID]	list low-level information on an object
docker events [CONTAINER]	list real time events from a container
docker port [CONTAINER]	show port (or specific) mapping from a container
docker top [CONTAINER]	show running processes in a container
docker stats [CONTAINER]	show live resource usage statistics of containers
docker diff [CONTINAER]	show changes to files (or directories) on a filesystem
docker images ls	show all locally stored images
docker history [IMAGE]	show history of an image

[5]

Full cheat sheet

Next page reference: [6]

1. Containers

A lightweight virtual OS that run processes in full isolation.

1.1 Lifecycle

- docker create creates a container but does not start it.
- docker rename allows the container to be renamed.
- docker run creates and starts a container in one operation.
- docker rm deletes a container.
- docker update updates a container's resource limits.
- docker run --rm : remove the container after it stops.
- docker run -v $HOSTDIR:$DOCKERDIR: map the directory ($HOSTDIR) on the host to a
- docker container ($DOCKERDIR).
- docker rm –v: remove the volumes associated with the container.
- docker run --log-driver=syslog : run docker with a custom log driver.

1.2 Starting and Stopping

- docker start starts a container so it is running.
- docker stop stops a running container.
- docker restart stops and starts a container.
- docker pause pauses a running container, "freezing" it in place.
- docker unpause will unpause a running
- container.
- docker wait blocks until running container
- stops.

docker kill sends a SIGKILL to a running container.

docker attach will connect to a running container.

1.3 CPU Constraints

CPU can be limited either using a percentage over all CPUs, or by using specific cores.

- -c or cpu-shares: 1024 means 100% of the CPU, so if we want the container to take 50% of all CPU cores, we should specify 512 for instance, docker run -ti --c 512 ...cpuset-cpus
- : use only some CPU cores, for instance, docker run -ti --cpuset-cpus=0,4,6 ...

1.4 Memory Constraints

Memory can be limited using –m flag, for instance, docker run -it -m 300M ubuntu:14.04 /bin/bash

1.5 Capabilities

cap-add and cap-drop: Add or drop linux capabilities.

- Mount a FUSE based filesystem:
  - docker run --rm -it --cap-add SYS_ADMIN --device /dev/fuse sshfs
- Give access to a single device:
  - docker run -it --device=/dev/ttyUSB0 debian bash
- Give access to all devices:
  - docker run -it --privileged -v /dev/bus/usb:/dev/bus/usb debian bash

1.6 Info

- docker ps shows running containers.
- docker logs gets logs from container. (You can use a custom log driver, but logs is only available for json-fileand journald in 1.10).
- docker inspect looks at all the info on a container (including IP address).
- docker events gets events from container.
- docker port shows public facing port of container.
- docker top shows running processes in container.
- docker stats shows containers' resource usage statistics.
- docker diff shows changed files in the container's FS.
- docker ps –ashows running and stopped containers

1.7 Import / Export

- docker cp copies files or folders between a container and the local filesystem.
- docker export turns container filesystem into tarball archive stream to STDOUT.

1.8 Executing Commands

docker exec to execute a command in container.

2. Images

A template or blueprint for docker containers.

2.1 Lifecycle

- docker images shows all images.
- docker import creates an image from a tarball.
- docker build creates image from Dockerfile.
- docker commit creates image from a container, pausing it temporarily if it is running.
- docker rmi removes an image.
- docker load loads an image from a tar archive as STDIN, including images and tags (as of 0.7).
- docker save saves an image to a tar archive stream to STDOUT with all parent layers, tags & versions (as of 0.7).

2.2. Info

- docker history shows history of image.
- docker tag tags an image to a name (local or registry).

2.3. Cleaning up

- docker rmi remove specific images.
- docker-gc a toolto clean up images that are no longer used by any containers in a safe manner.

2.4. Load/Save image

- docker load < my_image.tar.gz load an image from file
- docker save my_image:my_tag | gzip > my_image.tar.gz save an existing image

2.5. Import/Export container

- cat my_container.tar.gz | docker import - my_image:my_tag import a container as an image from file
- docker export my_container | gzip > my_container.tar.gz export an existing container

3. Networks

A small def goes here

3.1. Lifecycle

- docker network create
- docker network rm

3.2. Info

- docker network ls
- docker network inspect

3.3. Connection

- docker network connect
- docker network disconnect

4. Registry & Repository

A repository is a hosted collection of tagged images that together create the file system for a container.
A registry is a host -- a server that stores repositories and provides an HTTP API for managing the uploading and downloading of repositories.
Docker.com hosts its own index to a central registry which contains a large number of repositories.

- docker login to login to a registry.
- docker logout to logout from a registry.
- docker search searches registry for image.
- docker pull pulls an image from registry to local machine.
- docker push pushes an image to the registry from local machine.

5. Volumes

Docker volumes are free-floating filesystems. They don't have to be connected to a particular container. You should use volumes mounted from data-only containers for portability.

5.1. Lifecycle

- docker volume create
- docker volume rm

5.2. Info

- docker volume ls
- docker volume inspect

6. Exposing ports

- docker run -p 127.0.0.1:$HOSTPORT:$CONTAINER-
- PORT --name CONTAINER -t docker_image mapping the container port to the host port using –p
- EXPOSE <CONTAINERPORT>expose port CONTAINERPORT at runtime (see dockerfile)
- docker port CONTAINER $CONTAINERPORT check the mapped port

7. Tips

7.1. Get IP address

> docker inspect some_docker_id | grep IPAddress | cut -d '"' -f 4
or install jq:
> docker inspect some_docker_id | jq -r '.[0].NetworkSettings.IPAddress'
or using a go template:
> docker inspect -f '{{ .NetworkSettings.IPAddress }}' <container_name>

7.2. Get port mapping

```
docker inspect -f '{{range $p, $conf := .NetworkSettings.Ports}} {{$p}} -> {{(index $conf 0).HostPort}} {{end}}' <containername>
```

7.3. Find containers by regular expression

```
for i in $(docker ps -a | grep "REGEXP_PATTERN" | cut -f1 -d" "); do echo $i; done
```

7.4. Get Environment Settings

```
docker run --rm ubuntu env
```

7.5. Kill running containers

```
docker kill $(docker ps -q)
```

7.6. Delete old containers

```
docker ps -a | grep 'weeks ago' | awk '{print $1}' | xargs docker rm
```

7.7. Delete stopped containers

```
docker rm -v $(docker ps -a -q -f status=exited)
```

7.8. Delete dangling images

```
docker rmi $(docker images -q -f dangling=true)
```

7.9. Delete all images

```
docker rmi $(docker images -q)
```

7.10. Delete dangling volumes

```
docker volume rm $(docker volume ls -q -f dangling=true)
```

Exercises

Tiny project

Single-responsibility principle

Why we should decompose this project into tiny-tiny parts?

Wikipedia:

The single-responsibility principle (SRP) is a computer-programming principle that states that every class in a computer program should have responsibility over a single part of that program's functionality, which it should encapsulate. All of that module, class or function's services should be narrowly aligned with that responsibility [7].

Security

References

References I

- *What is a Container?* URL: https://www.docker.com/resources/what-container.
- *packhelp*. URL: https://packhelp.com/plain-shipping-box/.
- *64-bit memory layout*. URL: https://www.berthon.eu/wiki/foss:wikishelf:linux:memory.
- *Build a Docker Image just like how you would configure a VM*. URL: https://medium.com/platformer-blog/practical-guide-on-writing-a-dockerfile-for-your-application-89376f88b3b5.
- *List Of Docker Commands: Cheat Sheet*. URL: https://phoenixnap.com/kb/list-of-docker-commands-cheat-sheet.

References II

Docker Commands - Complete Cheat Sheet. URL: https://linoxide.com/linux-how-to/docker-commands-cheat-sheet/.

Single-responsibility principle. URL: https://en.wikipedia.org/wiki/Single-responsibility_principle.